

Reflective Journal

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In Module 04 Lab, we worked with Exploratory Data Analysis (EDA) using Python in Google Colab and explored the Titanic dataset. The goal of the lab was to understand how to examine and interpret data before building any machine learning models. This included identifying patterns, checking for missing values, and creating visualizations to better understand the dataset.

At the beginning of the lab, we use several imported Python libraries that are commonly used in data analysis. We imported pandas to load and organize the data into a table format that was easier to read and analyze. We used matplotlib.pyplot to create graphs and visualizations. We used seaborn, which is built on top of Matplotlib and makes it easier to create graphs that are more visually appealing and easier to understand. Using these libraries helped us see how different tools work together during data analysis.

One of the first functions we used was `.describe()`, which calculates descriptive statistics such as the mean, standard deviation, minimum, and maximum values for numerical columns. This helped our team quickly understand the dataset without looking at every individual row. For example, we learned that the average passenger age was about 29 years old, fares ranged from \$0 to over \$500, and only about 38% of passengers survived. We also use `.info()`, which showed us the structure of the dataset and revealed that some columns, such as Age and Cabin, had missing values. This helped us understand that real-world datasets are not always complete and that checking the data first is an important step.

For the visualizations, we used several different types of graphs. We used several different types of graphs. We used a countplot to visualize survival rates and saw that more passengers died than survived. Then, we used hue with countplots to compare survival by categories such as passenger class and gender. These graphs showed that first-class passengers survived more often than third-class passengers and that females had higher survival rates than males. We also used Seaborn's FacetGrid with `plt.hist()` to create histograms that displayed age distributions and survival outcomes. In addition, we created a countplot for Embarked to see whether the port of embarkation had any relationship with survival. Finally, we used a violin plot to compare Fare and Survived, which helped us see whether passengers who paid higher fares appeared more likely to survive.

As a team, this lab helped us realize that coding is more than just getting the code to run. Each function and graph had a purpose and helped us answer questions about the data. Since we are still learning how to code, this assignment made us more comfortable with Python syntax and helped us understand how libraries like Pandas, Matplotlib, and Seaborn can be used to explore and interpret data. Overall, this lab increased our confidence with Python and showed us why exploratory data analysis is an important first step before building machine learning models.